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JAPAN VAM & POVAL CO.,LTD.

SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name: J-POVAL **Polyvinyl Alcohol V-type**
VP-18PE, VP-20PE

Manufacturer

Company Name: **JAPAN VAM & POVAL CO., LTD.**
Address: 3-11-1, Chikko Shinmachi,
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9:00am – 5:00pm (Japan Time)

2. HAZARDS IDENTIFICATION.

〔GHS classification〕 It doesn't correspond to a standard for GHS classification.

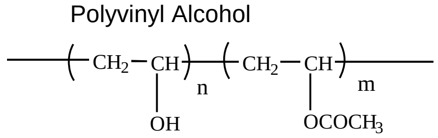
May be harmful if inhaled and ingested.
May cause eye and skin irritation.
Dust explosion may be caused.

NFPA : HR= health-0, flame-2, react-0

3. COMPOSITION / INFORMATION ON INGREDIENTS

Chemical Description: Polyvinyl Alcohol

Formula:



Composition / information on ingredients:

Polyvinyl Alcohol 97%

CAS Registry No.:	9002-89-5 (Fully Hydrolyzed),	25213-24-5 (Partially Hydrolyzed)
TSCA (USA):	On Inventory	
EINECS (EU):	209-183-3 (Fully Hydrolyzed),	209-183-3 + 203-545-4(Partially Hydrolyzed)
DSL (CANADA):	On Inventory	
AIIC (AUSTRALIA):	On Inventory	
ECL (SOUTH KOREA):	KE-29060 (Fully Hydrolyzed),	KE-00041 (Partially Hydrolyzed)
IECSC (CHINA):	On Inventory	
METI (JAPAN):	6-682	

Sodium Acetate 0.4%

CAS Registry No.:	127-09-3
TSCA (USA):	On Inventory
EINECS (EU):	204-823-8
DSL (CANADA):	On Inventory
AIIC (AUSTRALIA):	On Inventory
ECL (SOUTH KOREA):	KE-00061
IECSC (CHINA):	On Inventory
METI (JAPAN):	2-692

Water 2.6%

4. FIRST AID MEASURES

Inhalation: No special measures are necessary. Remove to fresh air.

Eye Contact: Immediately rinse eyes thoroughly, including under eyelids, with running water for at least 15 minutes.

Skin Contact: Remove contaminated clothes and shoes, rinse skin with plenty of water.

Ingestion: Emit by drinking water of room temperature. If poisoning occurs, contact a doctor or poisons information center.

5. FIRE FIGHTING MEASURES

Flammable Properties:

Flashpoint and Method:	above 100℃
Flammable Limits:	Not applicable
Auto Ignition Temperature:	440℃
Combustibility:	Low
Special Fire and Explosion Hazard:	Dust form explosive mixture with air. (120mj / (35g / m^3))

Extinguishing media: Water, Powder extinguisher, CO_2, Chemical foam

6. ACCIDENTAL RELEASE MEASURES

Land Spill: Sweep up and collect for disposal, remove with a shovel or any suitable equipment. Collect in a convenient manner to avoid dusty conditions.

Water Spill: When solution is abundant, it collects. It discharges, after carrying out activated sludge processing, when it mixes in waste water.

7. HANDLING AND STORAGE

Handling: Avoid contact with eyes, skin and clothing. Handle material with suitable protection.

Storage: Keep dry (Avoid moisture and high temperature)
Store away from sunlight in well ventilated dry place at room temperature.

8. EXPOSURE CONTOROLS / PERSONAL PROTECTION

Engineering Measure: Installation of partial exhaust equipment is desirable.

Personal Protective Equipment:

Respiratory protection: Respirator or Dust mask
Eye/Face protection: Safety Glasses (Goggles)
Skin Protection: Safety gloves and Protective clothing.

9. PHISICAL AND CHEMICAL PROPERTIES

Appearance:	Powder or Granular
Color:	White or Creamy
Odor:	Mild
Boiling Point :	Not applicable
Melting Point :	Over 200℃
Decomposition Point :	Over 200～250℃
Vapor Pressure:	Not applicable
Vapor Density:	Not applicable
pH:	5～7 (at 4wt% water solution)

Solubility in Water: Soluble in hot water, slightly soluble in cold water.
Temperatures of 90～95℃ are generally required for complete solution.

Specific Gravity : 1.19～1.31

Flammable Properties: Please refer to “5. FIRE FIGHTING MEASURES”

10. STABILITY AND REACTIVITY

Chemical Stability: Stable

Conditions to Avoid: Avoid moisture and high temperature

Hazardous Decomposition Products:
Hazardous gases produced are carbon monoxide from partial combustion

Dust explosiveness:

Product type	Granular PVA (GP type) Fine powder*1 content：5 % or less	Powder PVA (POW type) Fine powder*1 content：15~20 %	Ultra fine powder*2 PVA
Lower explosive limit: Determined according to JIS-Z8818 using dust explosion testing apparatus (tube type)	1000 mg/L or more	180～310 mg/L	35～40 mg/L
Minimum ignition energy: Determined according to VDI2263 and IEC61241-2-3	1000 mJ or more	500～1000 mJ or more	60～423 mJ

*1 Fine powder： #100 mesh pass (particle size under 150 μm)
*2 Ultra fine powder： #200 mesh pass (particle size under 75 μm)

The ability to cause dust explosion increases as the particle diameter decreases; therefore, adequate measures against dust explosion are required.

The dust explosiveness of PVA in granule form (GP type) is not more than the lower limit of measurement; however, depending on the fine powder content, it does not necessarily mean the dust explosion may not occur.

11. TOXICOLOGICAL INFORMATION

Polyvinyl Alcohol is safe and non-toxic material when properly handled. Feeding studies on laboratory animals have indicated a very low order of oral toxicity. Since all Polyvinyl Alcohol are made as technical quality, however, they are not recommended for inclusion in any food or other preparation which might be taken internally. Polyvinyl Alcohol is primarily not a skin irritant and does not produce skin contact sensization.

Acute Toxicity: Oral LD50 (rat) >2,000mg/kg
Dermal LD50 (rat) >2,000mg/kg
(IRI PROJECT No. 552896,552901 1992)

Chronic Exposure: Not available

NTP: Not listed
IARC: Not classifiable as to Carcinogenicity to Humans (Group 3)
OSHA: Not listed
ACGIH: Not listed

12. ECOLOGICAL INFORMATION

Ecotoxicity:

Fish Toxicity:
LC50>1,000mg/L over 48hours of exposure, fully hydrolyzed Polyvinyl Alcohol (investigated in killifishes)
Inhabitation of activity of waste water/bacteria in acclimated domestic slug microorganisms:
In such situations, Polyvinyl Alcohol present does not inhibit the metabolism of existing activated slug microorganisms. Activated slug microorganisms can be acclimated to Polyvinyl Alcohol under conditions attainable in conventional waste water treatment systems.

Environmental fate:

Chemical Oxygen Demand: 60～100 (mg/kg) $\times 10^4$ (JIS K-0102)
Biological Elimination: Polyvinyl Alcohol has a negligible biochemical oxygen demand.

13. DISPOSAL CONSIDERATIONS

Comply with all national and local regulations.
Do not dump this material into sewers, on the ground or into any body of water.
In accordance with government regulations for the disposal of special waste.
Incinerate in an approved incinerator, according to government regulations.
Dispose of waste disposal contractor to a regulated landfill.

14. TRANSPORT INFORMATION

IATA: Not Restricted
UN classification No.: Not applicable
IMDG classification: Not applicable

15. REGULATORY INFORMATION

US Regulations;

NTP Test. Prog.
EPA: CERCLA PQ=Not listed
: EPCRA TPQ=Not listed
OSHA: TQ=Not listed

EU Regulations;

SYMBOL: -----------
R-phrases: ----------
S-phrases: ----------
EC index No.: Not listed

16. OTHER INFORMATION

No specific notes

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